SECTION B

QUESTION 6

(a) Factorise fully:

(1) $x^2 - \frac{9}{x^2}$ (2)

FE: if drop denom.

$= \left(x + \frac{3}{x}\right)\left(x - \frac{3}{x}\right)$ ✓ ✓ or $\frac{1}{x^2}(x^2+3)(x^2-3)$

(2) $4x^2 - 8x + 2a - ax$ (4)

$= 4x(x-2) + a(2-x)$ ✓ ✓

$= 4x(x-2) - a(x-2)$ ✓

$= (x-2)(4x-a)$ ✓

(b) Simplify: $\frac{3x^2 - 6x + 12}{x^3 + 8}$ (4)

$= \frac{3(x^2 - 2x + 4)}{(x+2)(x^2 - 2x + 4)}$ ✓ HCF ✓ ✓

$= \frac{3}{x+2}$ ✓

(c) Simplify the following expression: $\frac{2.3^x - 4.3^{x-2}}{3^x - 3^{x-1}}$ (5)

$= \frac{2.3^x - 2^2.3^x.3^{-2}}{3^x - 3^x.3^{-1}}$ ✓ split bases ✓

BD → expn. laws → ÷ over ±

$= \frac{3^x\left(2 - \frac{4}{9}\right)}{3^x\left(1 - \frac{1}{3}\right)}$ ✓ factorise

$= \frac{14}{9} \times \frac{3}{2}$ ✓

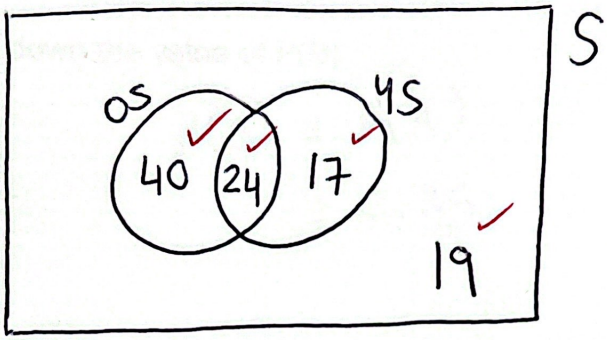
$= \frac{7}{3}$ ✓ ca

[15]

QUESTION 9

A survey of 100 pupils shows that 64 pupils have an older sibling. 41 pupils have a younger sibling and 24 pupils have both. (A sibling is a brother or sister)

(a) Draw a Venn diagram illustrating the results of the survey. (4)



(b) Determine the probability (as a simplified fraction) that a pupil chosen at random has:

(1) an older sibling (1)

$\frac{64}{100} = \frac{16}{25}$ ✓

(2) only a younger sibling (1)

$\frac{17}{100}$ ✓

(3) no siblings (1)

$\frac{19}{100}$ ✓

(4) an older or a younger sibling (1)

$\frac{81}{100}$ ✓

(c) Write down the equation of line AC. (1)

$x = 4$ ✓ca

(d) Write down the equation of line BC. (1)

$y = -4$ ✓ca

(e) Write down the length of AC. (1)

AC = 8 units ✓ca

(f) The line through A and B is an axes of symmetry of *f*. Write down the equation of the other axis of symmetry of *f*. (1)

$y = -x$ ✓

(g) Write down the equation of the graph of *f*, shifted down 2 units. (1)

$y = \frac{16}{x} - 2$ ✓ca

[12]

QUESTION 5

(a) Consider the linear sequence: 5; 8; 11; ...

(1) Determine the nth term of the sequence. (2)

$T_n = 3n + 2$ ✓ ✓

(2) Calculate the value of the 15th term of the sequence. (2)

$T_{15} = 3(15) + 2$ ✓

$= 47$ ✓ca

(3) Which term of the sequence is equal to 83? (2)

$83 = 3n + 2$ ✓

$81 = 3n$

$\therefore n = 27^{th}$ term ✓

(b) A linear number pattern is represented by:

$x + 3; 3x + 2; 6x - 1.$

Determine the numerical value of x. (4)

$3x + 2 - (x + 3) = 6x - 1 - (3x + 2)$ ✓ ✓

$3x + 2 - x - 3 = 6x - 1 - 3x - 2$

$2x - 1 = 3x - 3$ ✓m

$2 = x$ ✓

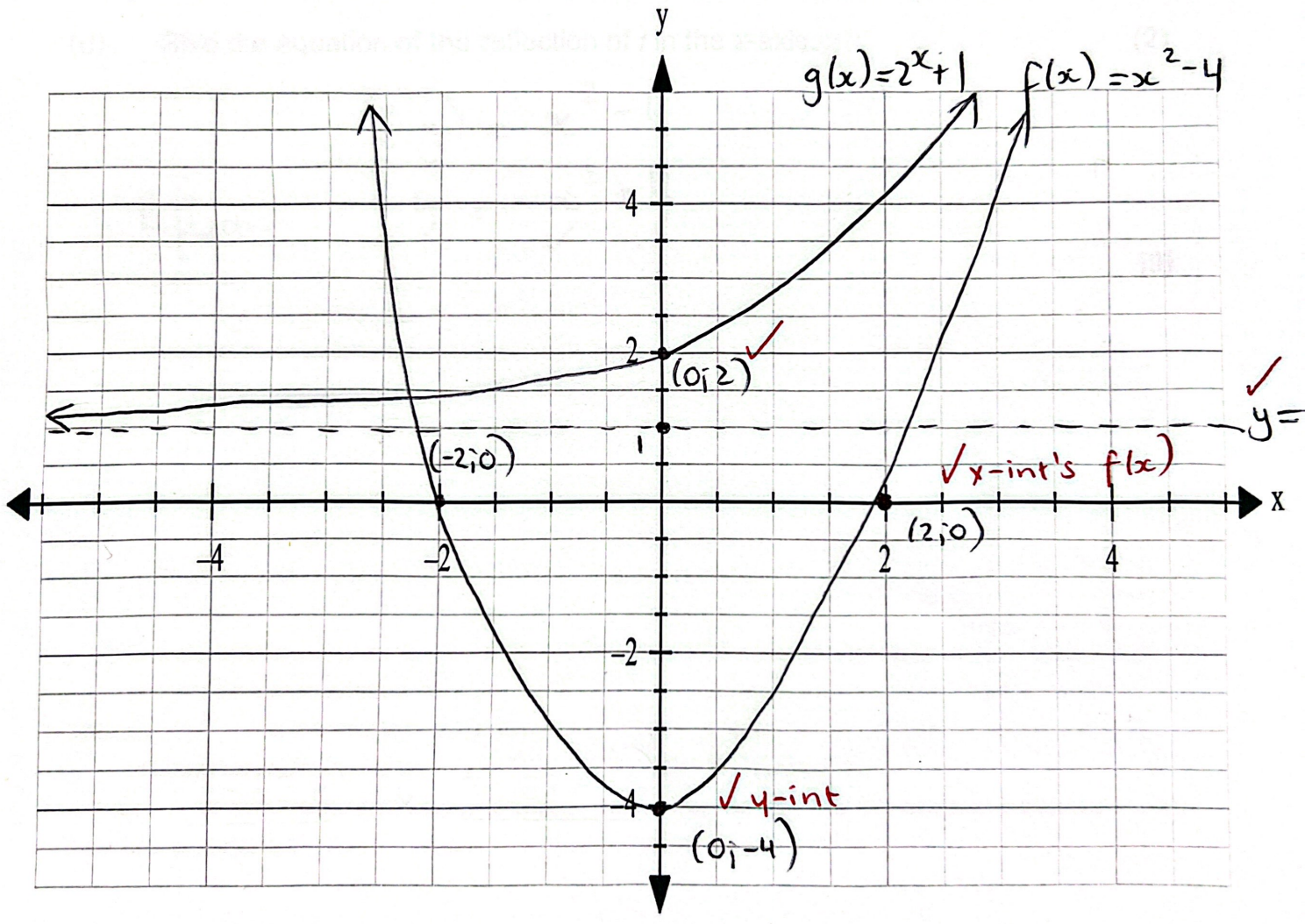
[10]

SECTION A: 50

QUESTION 4

Given functions $f(x) = x^2 - 4$ and $g(x) = 2^x + 1$.

(a) Draw f and g on the same system of axes. Clearly indicate all intercepts with the axes, turning points and/or asymptotes. (4)



QUESTION 2

Solve for x in the following equations:

(a) $(5^x)^6 = 125$ (3)

$5^{6x} = 5^3$ ✓remove bracket/same bases.

$6x = 3$ ✓equate exp.

$\therefore x = \frac{1}{2}$ ✓ca

(b) $(x-8)(x+4) = 45$ (4)

$x^2 - 4x - 32 = 45$ ✓

$x^2 - 4x - 77 = 0$ ✓

$(x-11)(x+7) = 0$ ✓

$x = 11$ or $x = -7$ ✓ca both

(c) $\frac{x+7}{x-5} + \frac{x+5}{x-7} = 2$ (6)

LCD: $(x-5)(x-7)$ ✓

$(x+7)(x-7) + (x+5)(x-5) = 2(x-5)(x-7)$

$x^2 - 49$ ✓ $+ x^2 - 25$ ✓ $= 2(x^2 - 12x + 35)$ ✓

$2x^2 - 74 = 2x^2 - 24x + 70$

$24x = 144$ ✓m

$x = 6$ ✓

GRADE 10 FORMULA SHEET:

$$A = P(1 + in)$$

$$A = P(1 + i)^n$$

$$y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

De La Salle Holy Cross College

NOVEMBER EXAMINATIONS

2024

Grade 10

MATHEMATICS

Paper 1

Examiner: Mrs Seewald **Time:** 2 hours

Moderators: Mrs De Freitas & Mr Guilhermino **Marks:** 100

Name: MEMO (Final Draft)			
Teacher:	GDF	AS	PG

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 15 pages that include 1 page for extra calculations or rough work. Please check that your question paper is complete.
2. Answer all the questions in the space provided on the question paper.
3. A non-graphical, non-programmable calculator may be used.
4. Ensure that your calculator is set in degree mode.
5. All necessary calculations must be shown clearly. Answers only will not necessarily be awarded full marks.
6. Diagrams and graphs are not necessarily drawn to scale.
7. Round off final answers to **ONE** decimal place, unless stated otherwise.
8. It is in your own interest to write legibly and to present your work neatly.
9. One blank page (page 15) is included in the paper. If you run out of space for a question, then use this page. Clearly indicate the question number of your answer should you use this extra space.

	Section A					**Section B**					**Total**
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
Mark											
Total	8	16	7	9	10	15	12	7	11	5	**100**

GdF PG AS

Remember that God is with you and will help and inspire you in all that you do

SECTION A

QUESTION 1

(a) Given: $q = \sqrt{b^2 - 4ac}$

(1) Determine the value of q if $a = 2$, $b = -1$ and $c = -4$. (2)

$q = \sqrt{(-1)^2 - 4(2)(-4)}$ ✓m sub

$= \sqrt{1 + 32}$ ✓

$= \sqrt{33}$ or 5,7

(2) State whether your answer for q is rational or irrational. (1)

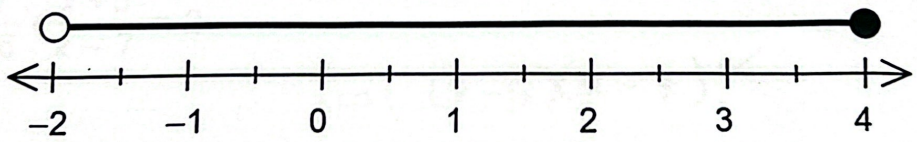
Irrational. ✓

(3) Between which TWO consecutive integers does q lie? (1)

$5 < q < 6$ ✓ both correct

(b) Answer the following questions regarding representation of numbers.

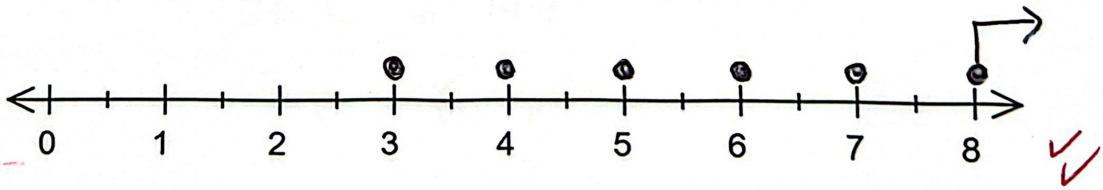
(1) Rewrite the following in interval notation: (2)



$x \in (-2; 4]$ ✓ ✓

(2) Represent the following on the number line provided below: (2)

$\{x : x \geq 3;\ x \in \mathbb{N}_0\}$



✓ ✓

[8]

(d) $-3 \leq \frac{2x-1}{3} < 1$ (3)

$-9 \leq 2x-1 < 3$ ✓×3

$-8 \leq 2x < 4$ ✓ADD

$-4 \leq x < 2$ ✓÷2

[16]

QUESTION 3

Simplify fully:

(a) $(x - 1 - x^2)(3 - x)$ (3)

$= 3x - 3 - 3x^2 - x^2 + x + x^3$ ✓ ✓

$= -4x^2 + 4x - 3 + x^3$ ✓ca

(b) $\frac{3x-9y}{2x+5} \div \frac{3}{4x+10}$ (4)

$= \frac{3(x-3y)}{(2x+5)} \times \frac{2(2x+5)}{3}$ ✓fact ✓reflc ✓fact

$= 2(x-3y)$ or $2x-6y$ ✓ca

[7]

(b) Determine the range of g. (1)

$y > 1$ ✓ or $y \in (1; \infty)$

(c) For which value(s) of x is $f(x) > 0$. (2)

$x < -2$ ✓ or $x > 2$ ✓

(d) Give the equation of the reflection of f in the x-axis. (2)

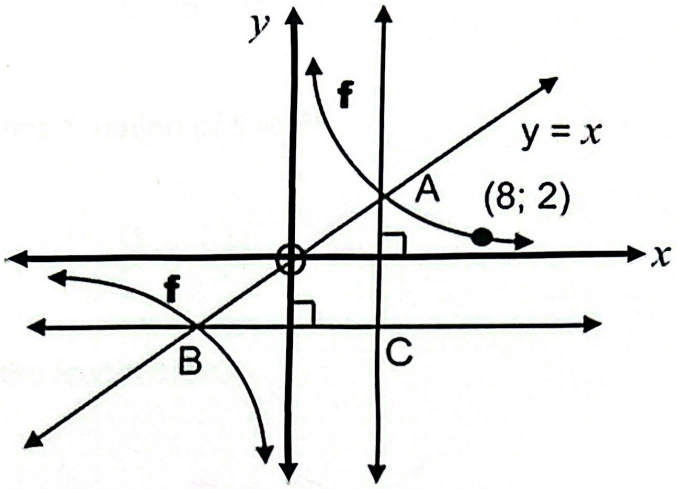
$(-y) = x^2 - 4$

Reflection: $y = -x^2 + 4$ ✓ ✓

[9]

QUESTION 7

The diagram of $y = x$ and function f is given below.



(a) Write down the equation of f. (2)

$y = \frac{q}{x} + q$

$y = \frac{q}{x}$

$\therefore \quad 2 = \frac{q}{8}$ $\qquad \therefore \quad a = 16$

$q = 0$ ✓

$\therefore f(x) = \frac{16}{x}$ $\quad a = 16$ ✓

(b) Calculate the coordinates A, B and C. (5)

$y = x$ $\qquad y = \frac{16}{x}$

$x = \frac{16}{x}$

$x^2 = 16$

$\therefore x = \pm 4$

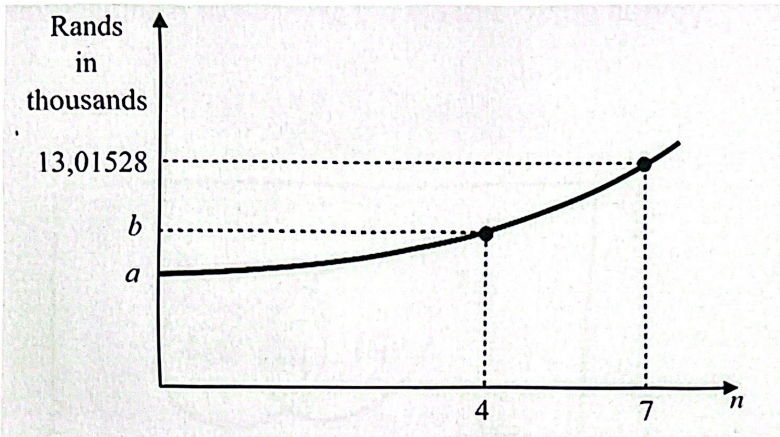
$\therefore$ A $(4; 4)$ ✓ ✓

$\therefore$ B $(-4; -4)$ ✓ ✓

$\therefore$ C $(4; -4)$ ✓ca

QUESTION 8

The following graph shows the growth of an amount of money invested for n years. The interest rate remains 7,2% p.a. throughout the whole period:



Source: Mind Action Series Textbook

(a) Which type of interest applies in this scenario? (1)

Compound Interest ✓

(b) What does the number *a* represent? (1)

Original / starting amount that is invested. ✓

(c) Calculate, the value of:

(1) *a* (3)

$A = P(1+i)^n$ subtract 1 if incorrect formula used

$1301528 \times 10^3 = a(1 + 7{,}2\%)^7$ ✓ ✓

$R\,8000 = a$ ✓a

(2) *b* (2)

$A = 8000\left(1 + \frac{7{,}2}{100}\right)^4$ ✓m

$= 10\,564{,}99$ ✓a

[7]

(c) Two events, A and B, are complementary and exhaustive. Also,

$P(\overline{A}) = 0{,}35$.

(1) Complete the statement: $P(A) + P(B) = ..1.$ ✓ (1)

(2) Write down the value of P(A and B). (1)

0 ✓

(3) Write down the value of P(B). (1)

$P(B) = P(\overline{A})$ ✓

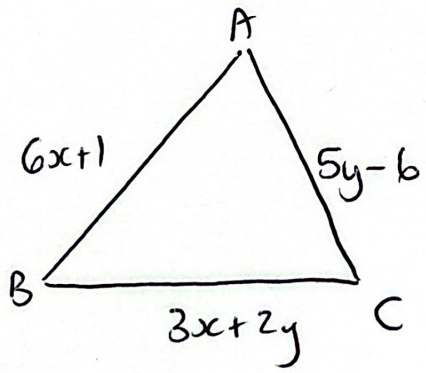
$= 0{,}35$ or $\frac{7}{20}$

[11]

QUESTION 10

In Δ ABC, $AB = 6x + 1$, $BC = 3x + 2y$ and $AC = 5y - 6$.

Determine the value of x and y if Δ ABC is equilateral.



$\therefore 6x + 1 = 3x + 2y$

$3x = 2y - 1$

$\therefore 3x = 2(x + 2) - 1$ ✓sub

$3x = 2x + 4 - 1$

$x = 3$ ✓ →

$\therefore y = 3 + 2$ ✓sub

$y = 5$ ✓ →

$5y - 6 = 3x + 2y$

$3y = 3x + 6$

$y = x + 2$ ✓

[5]

SECTION B: 50

TOTAL: 100